

Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server stores essential business information, including customer and operational data that supports critical decision making. Securing this data is vital to maintaining business continuity, protecting sensitive information, and maintaining customer trust. If the server were compromised or disabled, core business functions could be disrupted, resulting in operational delays, data loss, and financial impact. This analysis evaluates risks to strengthen the system's security posture.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtain sensitive information via exfiltration	3	3	9
Employee	Disrupt mission critical operations	2	3	6
Customer	Alter/Delete critical information	1	3	3

Approach

This assessment used a qualitative approach based on security expertise, likelihood estimation, and business impact considerations. Threat sources were selected due to their access to the system, capability to cause harm, and relevance to the organization's environment. Each threat event reflects realistic risks that could disrupt business operations or expose sensitive data. Assessing these scenarios helps stakeholders understand the most critical areas requiring security attention.

Remediation Strategy

To mitigate the identified risks, multiple security controls should be implemented. Applying the principle of least privilege and role based access control limits user capabilities and reduces opportunities for misuse. Multi factor authentication, strong passwords, and network allow listing strengthen authentication and prevent unauthorized access. Encryption using TLS ensures data transmitted across the network remains protected. Implementing layered defenses and auditing mechanisms provides continuous monitoring and reduces the likelihood and impact of attacks.